CSE 165 Project 3 — Design Document

Project summary

This project implements a gesture-aware embodied agent for Meta Quest 2 mixed reality. The user scans the room, points at a reachable floor destination with the right hand, confirms the command with a left-hand pinch, and watches the animated agent move through the captured room while respecting anchored wall proxies.

Implemented system

- Passthrough-ready mixed-reality scene with room-surface detection.
- Anchored floor and wall proxies that act as digital copies of the physical space.
- Joint-based gesture recognition using right-hand pointing plus left-hand pinch confirmation.
- Mixamo-exported Beetlejuice character with body/head textures reapplied in Unity; the agent turns, walks, arrives, and stops when a wall proxy blocks the route.
- In-headset status readout for captured surfaces, gesture state, and navigation state.

Design exploration

Implemented concept: point + pinch confirmation

The user points with the right index finger to choose a floor destination. A visible reticle previews the selected point. The user then pinches the left thumb and index finger to commit the command, after which the embodied agent turns and walks toward the destination unless an anchored room surface blocks the route.

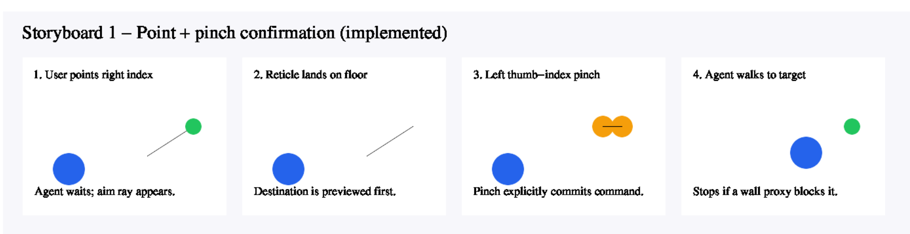

Figure 1: Point + pinch storyboard

Alternative concept: directional swipe

The user raises an open palm, swipes in a direction, and the system converts the motion vector into a movement command. This is fast and expressive, but it is easier to trigger accidentally than a deliberate point-and-confirm gesture.

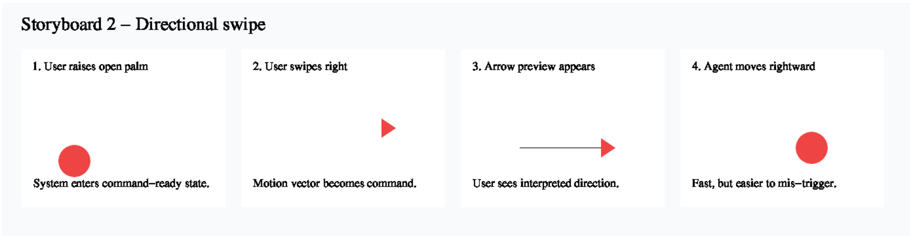

Figure 2: Directional swipe storyboard

Alternative concept: two-hand vector

The left hand establishes an origin while the right hand defines a direction vector. Releasing the pose commits the command. This is precise and camera-independent, but it asks more of the user than the implemented method.

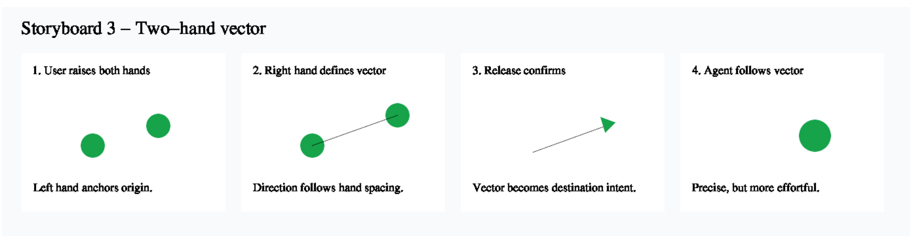

Figure 3: Two-hand vector storyboard

Chosen interaction rationale

Point + pinch was selected because it separates *preview* from *commit*. That makes the user's intention legible, keeps the command spatially grounded in the real room, and reduces accidental movement while still relying only on tracked hand joints.

Heuristic evaluation

Heuristic	Violation + rationale	Severity	Recommendation
Visibility of system status	Mostly satisfied. The reticle, aim line, and status panel reveal pointing state, captured floor/wall counts, and agent state, but newly detected surfaces still require a moment of interpretation.	Low	Keep the status panel visible during scanning and narrate the translucent surface colors in the demo.
Match between system and real world	Partially satisfied. Pointing at a physical floor is intuitive, but a detected wall proxy can be imperfect if the room scan is incomplete.	Medium	Let the user rescan or manually confirm questionable wall proxies before recording the final demo.
User control and freedom	Partially satisfied. A second destination command redirects the agent, but there is no dedicated cancel/stop gesture once walking begins.	Medium	Add an open-palm stop gesture if time permits.

Heuristic	Violation + rationale	Severity	Recommendation
Consistency and standards	Mostly satisfied. Right hand selects, left hand confirms, and the same reticle semantics persist throughout the experience. The fixed handedness assumption may not fit every user.	Low	Add a handedness toggle for left-handed users.
Error prevention	Mostly satisfied. Pinch confirmation prevents accidental motion and wall proxies stop the agent before collision, but missing room scans can still leave unseen obstacles.	Medium	Require at least one floor and one wall capture before the final demonstration sequence.
Recognition rather than recall	Mostly satisfied. The visible reticle and status text reduce memory burden, but first-time users still need to learn the left-hand pinch trigger.	Low	Show a short onboarding placard before free interaction.

Heuristic	Violation + rationale	Severity	Recommendation
Flexibility and efficiency of use	Satisfied for the assignment scope. The selected gesture is quick after learning and the editor fallback accelerates iteration without changing the Quest interaction.	Low	If the project grows later, add optional waypoint queuing rather than overloading the current gesture.
Aesthetic and minimalist design	Mostly satisfied. Translucent room proxies keep the real environment visible, though hand-joint debug markers are visually busy once calibration is complete.	Low	Hide debug markers for the polished recording after proving joint access.
Help users recognize, diagnose, and recover from errors	Partially satisfied. The status panel reports "blocked by room surface," but the exact blocking wall is not highlighted.	Medium	Flash the blocking proxy when the agent stops.

Heuristic	Violation + rationale	Severity	Recommendation
Help and documentation	Partially satisfied. The README, checklist, and status panel explain the system, but there is no in-headset first-run tutorial.	Medium	Add a concise onboarding card if there is time before submission.

Submission notes

- The interaction implemented for the final demo is **point + pinch confirmation**.
- The strongest grading sequence is: show passthrough, scan a floor and wall, show tracked hand joints, point and pinch to move the agent, then command a destination behind a wall proxy so the agent visibly stops.
- The repository includes the required storyboards and heuristic evaluation; final submission still needs the public GitHub link and headset-recorded video.